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CHAPTER 13

FUEL-OIL PIPING AND STORAGE

SECTION MC 1301 GENERAL

1301.1 Scope. This chapter shall govern the design, installation, construction and repair of fuel-oil storage and piping systems. The storage of flammable and combustible liquids not addressed in this chapter shall be in accordance with the *New York City Fire Code*.

1301.2 Storage and piping systems. Fuel-oil storage and piping systems shall comply with the requirements of this chapter, Appendix C and, to the extent not otherwise provided for in this code, shall comply with the requirements of NFPA 31. All above-ground and underground storage facilities shall also comply with the requirements of the *New York State Department of Environmental Conservation's Petroleum Bulk Storage Code*.

1301.3 Fuel type. An appliance shall be designed for use with the type of fuel to which it will be connected. Such appliance shall not be converted from the fuel specified on the rating plate for use with a different fuel without conforming with its listing and manufacturers specifications and securing reapproval from the commissioner.

1301.4 Fuel tanks, piping and valves. The tank, piping and valves for appliances burning oil shall be installed in accordance with the requirements of this chapter. Where oil-burning appliances are served by a tank located such that any part of the tank is above the level of the burner inlet connection and where the fuel supply line is taken from the top of the tank, an approved anti-siphon valve or other siphon-breaking device shall be installed. The anti-siphon valve or siphon-breaking device shall be located at the highest point in the supply line.

Exceptions: An anti-siphon valve or other siphon-breaking device shall not be required where either:

1. An approved foot valve is used in the tank, or
2. No. 6 fuel oil is used.

1301.5 Tanks abandoned or removed. All exterior above-grade fill piping shall be removed when tanks are abandoned or removed. Tank abandonment and removal shall be in accordance with Section 3404 of the *New York City Fire Code*.

1301.6 Out of service system. Fuel-oil storage systems that are temporarily or permanently taken out of service shall comply with the requirements of the *New York City Fire Code* and of the *New York State Department of Environmental Conservation's Petroleum Bulk Storage Code*.

1301.7 Fuel-oil spill and overfill prevention equipment. Fuel-oil spill and overfill prevention equipment shall comply with EPA 40 CFR Parts 280, and Section 1305.6.6.

1301.8 Portable fire extinguishers. Portable fire extinguishers shall be provided as required by the *New York City Fire Code* and NFPA 10.

1301.9 Absorbent materials. The building owner shall maintain a sufficient quantity of absorbent materials near fuel-oil storage tanks, pumps, and related equipment to control leaks and slipping hazards.

1301.10 Certificate of fitness. Where fuel-oil piping systems utilize pumps to transfer fuel oil to equipment at levels above the lowest floor or to storage tanks at levels above the lowest floor in buildings, a qualified employee or contracted general company holding a certificate of fitness from the Fire Department shall maintain the fuel-oil system.

SECTION MC 1302 MATERIAL

1302.1 General. Piping materials shall conform to the ASTM standards cited in this section.

1302.2 Rated for system. All materials shall be rated for the operating temperatures and pressures of the system, and shall be compatible with the type of liquid being handled by the system.

1302.3 Pipe standards. Fuel-oil pipe shall comply with Sections 1302.3.1 and 1302.3.2.

1302.3.1 Fuel-oil pipe. Fuel-oil pipe shall comply with the standards listed in Table 1302.3.

1302.3.2 Systems serving appliances above ground above the lowest floor. Piping for fuel-oil systems utilizing a transfer pump to appliances at levels above ground above the lowest floor or to storage tanks at levels above the lowest floor in buildings shall comply with the requirements of Section 1305.9.5.

**TABLE 1302.3
FUEL-OIL PIPING**

MATERIAL	STANDARD (see Chapter 15)
Copper, brass, or copper-alloy pipe	ASTM B 42; ASTM B 43; ASTM B 302
Copper or copper-alloy tubing (Type K, or L (PVC coated))	ASTM B 75; ASTM B 88; ASTM B 280
Nonmetallic pipe	ASTM D 2996
Flexible pipe systems	(See Section 1302.9)
Steel pipe	ASTM A 53/A 53M; ASTM A 106
Steel tubing ^a	ASTM A 254

a. Steel tubing shall only be permitted when installed by the equipment manufacturer in accordance with UL 2200 and UL labeled.

1302.4 Nonmetallic pipe. Nonmetallic pipe shall be listed and labeled as being acceptable for the intended application for flammable and combustible liquids. Nonmetallic pipe shall be installed only outside, underground.

1302.5 Fittings and valves. Fittings and valves for the piping systems shall be compatible with, or shall be of the same material as, the pipe or tubing, and shall conform with Table 1302.5.

**TABLE 1302.5
PIPE FITTINGS**

MATERIAL	STANDARD
Copper, brass, and copper alloys	ASME B16.15; ASME B16.18; ASME B16.22; ASME B16.24
Labeled pipe	In accordance with the piping system label and Section 1302.4
Nonmetallic pipe fittings	In accordance with the piping system label and Section 1302.4

Flexible pipe systems	Fittings are allowed in the FO pump room and in the room where the appliance served is located. Intermediate fittings are not allowed inside the buildings.
Steel	ASME B16.5; ASME B16.9; ASME B16.11; ASME B16.28; ASTM A 53/A 53M; ASTM A 106; ASTM A 234; ASTM A 420

1302.6 Bending of pipe. Pipe shall be suitable for bending and shall conform with Table 1302.3. Pipe bends shall be made with approved equipment. The bend shall not exceed the structural limitations of the pipe.

1302.7 Pumps. Pumps that are not part of an appliance shall be of a positive-displacement type. The pump shall automatically shut off the supply when not in operation.

1302.8 Flexible connectors and hoses. Flexible metal connectors and hoses used where rigid connections are impractical or to reduce the effect of jarring and vibration shall be listed and labeled in accordance with UL 2039 and shall be installed in compliance with its label and the manufacturer's installation instructions and shall not exceed 18 inches (457.2 mm). Connectors made from combustible materials shall not be used inside buildings or above ground outside of buildings.

1302.9 Flexible fuel-oil piping systems with continuous vacuum leak detection. Flexible fuel-oil piping systems with continuous vacuum leak detection consisting of a single wall or double wall metallic primary carrier encased with outer polymer jacket shall comply with the requirements of Sections 1302.9.1 through 1302.9.2.

†††**1302.9.1 Listing.** Flexible fuel-oil piping systems with continuous leak detection shall be listed and labeled in accordance with UL 1369.

†††**1302.9.2 Design and installation.** Flexible fuel-oil piping systems with continuous leak detection shall comply with the requirements of Sections 1302.9.2.1 through 1302.9.2.6.

1302.9.2.1 Installation. Flexible fuel-oil piping systems shall be designed and installed in accordance with this code, manufacturer's recommendation, and their listing.

1302.9.2.2 Outdoor applications. Flexible fuel-oil piping systems with continuous vacuum leak detection may be used for above ground outdoor applications, if such systems are double metallic piping with polymer protective cover for exterior exposure to the elements. A fire-resistance-rated enclosure shall not be required for such applications.

1302.9.2.3 Continuous pipe run. Flexible fuel-oil piping systems shall be installed with a single continuous pipe run from the transfer pump to the appliance or storage tank it serves. Intermediate couplings are not allowed within buildings.

1302.9.2.4 Levels above the lowest floor. Where installed above ground, flexible fuel-oil piping systems connecting a transfer pump to equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall also comply with the requirements of Section 1305.9.

1302.9.2.5. Horizontal offsets. Horizontal offsets shall comply with Section 1305.9.3.

1302.9.2.6 Leak detection and alarm. Flexible fuel-oil piping systems shall be installed with continuous vacuum leak detection and alarm. The alarm shall sound and the transfer pump shall stop when a leak is detected. The alarm shall be connected to a local audible alarm and to a remote alarm located at a supervising station. The wiring shall comply with the *New York City Electrical Code*.

SECTION MC 1303 JOINTS AND CONNECTIONS

1303.1 General. Joints and connections shall conform to the ASTM Standards listed in Table 1302.5, shall be of a type approved for fuel-oil piping systems, shall be rated for the temperatures and pressures of the systems in which

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the devices are installed, and shall be compatible with the fluid and all materials used. All threaded joints and connections shall be made tight with suitable lubricant or pipe compound. Unions and flanges, right or left couplings, and sweat fittings shall be brazed in accordance with ASME *Boiler and Pressure Vessel Code*, Section IX (Welding and Brazing Qualifications) or in accordance with AWS B2.2 Standard for Brazing Procedure and Performance Qualification. Cast-iron fittings shall not be used. Joints and connections shall be tight for the pressure required by test. Flanged joints requiring gaskets or packing shall be equipped with gaskets rated for a minimum of 750°F (398.9°C).

1303.1.1 Joints between different piping materials. Joints between different piping materials shall be made with adapter fittings. Joints between different metallic piping materials shall be made with approved dielectric fittings. All such fittings shall conform with the requirements of Section 1203.

1303.2 Reserved.

1303.3 Joint preparation and installation. Where required by Sections 1303.4 through 1303.10, the preparation and installation of brazed, threaded and welded joints shall comply with Sections 1303.3.1 through 1303.3.4.

1303.3.1 Brazed joints. All joints shall be brazed in accordance with ASME *Boiler and Pressure Vessel Code*, Section IX Welding and Brazing Qualifications or in accordance with American Welding Society AWS B2.2 Standard for Brazing Procedure and Performance Qualification.

1303.3.2 Reserved.

1303.3.3 Threaded joints. Threads shall conform to ASME B1.20.1. Pipe-joint compound or tape shall be applied on the male threads only and shall be compatible for application on the piping material and fluid.

1303.3.4 Welded joints. All joints shall be welded in accordance with ASME *Boiler and Pressure Vessel Code*, Section IX Welding and Brazing Qualifications or in accordance with AWS B2.1 Specifications for Welding Procedure and Performance Qualification.

1303.4 Brass pipe. Joints between brass pipe or fittings shall be brazed, threaded or welded joints complying with Section 1303.3.

1303.5 Reserved.

1303.6 Copper or copper-alloy pipe. Joints between copper or copper-alloy pipe or fittings shall be brazed, threaded or welded joints complying with Section 1303.3.

1303.7 Copper or copper-alloy tubing. Joints between copper or copper-alloy tubing or fittings shall be brazed joints complying with Section 1303.3 or flared joints. Flared joints shall be made by a tool designed for that operation.

1303.8 Nonmetallic pipe. Joints between nonmetallic pipe or fittings shall be installed in accordance with the manufacturer's instructions for the labeled pipe and fittings.

1303.9 Steel pipe. Joints between steel pipe or fittings shall be threaded or welded joints complying with Section 1303.3.

1303.10 Reserved.

1303.11 Piping protection. Proper allowance shall be made for expansion, contraction, jarring and vibration. Piping other than tubing, connected to underground tanks, except straight fill lines and test wells, shall be arranged to permit the tanks to settle without impairing the tightness of the piping connections. Piping serving equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall also comply with the requirements of Sections 1305.9.6 and 1305.9.7. Piping that is routed outside of the building or on the roof shall be provided with a protective coating to prevent rust. Supports for such piping shall be made of stainless steel or other corrosion resistive material.

SECTION MC 1304 PIPING SUPPORT

1304.1 General. Pipe supports shall be in accordance with Section 305. Piping serving equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall also comply with the requirements of Sections 1305.9.6 and 1305.9.7.

SECTION MC 1305 FUEL-OIL SYSTEM INSTALLATION

1305.1 General. Fuel-oil piping systems shall be installed in accordance with this section.

1305.2 Protection of pipe, equipment and appliances. All fuel-oil pipe, equipment and appliances shall be protected from physical damage. Piping serving equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall also comply with the requirements of Section 1305.9.

1305.2.1 Flood hazard. All fuel-oil pipe, tanks, equipment and appliances located in flood hazard areas shall comply with Appendix G of the *New York City Building Code*.

1305.2.2 Identification of concealed fuel-oil piping. Where horizontal fuel-oil piping is concealed within the building construction, such piping at intervals not more than 40 feet (12 192 mm) in length and at all changes of direction shall be identified by a permanent label or tag. The label or tag shall be located outside of the enclosure.

1305.3 Piping. Piping shall comply with the requirements of Sections 1305.3.1 through 1305.3.9.

1305.3.1 Size. The fuel-oil system shall be sized for the maximum capacity of fuel oil required. The minimum size of a supply line shall be $\frac{3}{8}$ -inch (9.5 mm) inside diameter nominal pipe or $\frac{3}{8}$ -inch (9.5 mm) OD tubing.

1305.3.2 Connections to tank. Supply piping shall connect to the top of the fuel-oil tank.

Exception: Storage tanks in buildings that comply with all of the following conditions:

1. The tank is located above ground on the lowest floor;
2. The tank does not exceed 330 gallons (1249.2 L); and
3. The tank is provided with a $\frac{1}{2}$ -inch (12.7 mm) opening for gravity discharge in the bottom of the tank.

1305.3.3 Pumps. Fuel oil shall be supplied by a transfer pump or automatic pump.

1305.3.4 Safeties and heat detectors. Appropriate safeties shall be provided so that detection of heat within the generator or equipment room shall prevent additional fuel oil from being pumped into the piping system within such room, including fusible link operated valves in the supply pipe at the suction side of the fuel oil transfer pump and in the supply pipe entering the generator, fuel oil burning equipment room or rooftop equipment enclosure.

1305.3.5 Horizontal runouts. Horizontal runouts from risers to the generator or equipment room shall follow as direct a route as practicable.

1305.3.6 Supply piping. Supply piping from a transfer pump to appliances at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall also comply with the requirements of Section 1305.9.

1305.3.7 Return piping. In direct feed systems, the minimum size of a return line shall be no less than the size of the supply piping specified in Section 1305.3.1. Valves shall not be installed on return piping unless a means of relieving overpressure is provided. Return piping serving equipment at levels above the lowest floor in buildings shall also comply with the requirements of Section 1305.9.

1305.3.8 Overflow piping. In systems with fuel oil storage tanks above the lowest level, overflow piping from tanks installed above the lowest floor shall be piped by gravity into the top of the tank at the lowest floor that supplies fuel to such tank. The overflow pipe shall be minimum two pipe sizes larger than fuel oil supply pipe. Valves shall not be installed on overflow piping.

1305.3.9 Direct feed. Systems where day tanks are absent (such as generator installations where fuel oil is taken directly from a fuel-oil pipe or header into the engine) shall comply with Section 1305.9.12.

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1305.4 Reserved.

1305.5 System pressure. The system shall be designed for the maximum pressure required by the fuel-oil-burning appliance. Air or other gases shall not be used to pressurize tanks. Pressure in a storage tank for the purpose of discharging oil shall be prohibited.

1305.6 Fill piping. Fill piping shall comply with the requirements of Sections 1305.6.1 through 1305.6.6.

1305.6.1 Size. Fill piping shall be a minimum of 2 inches (50.8 mm) in diameter or 3 inches (76.2 mm) for No. 6 fuel oil.

1305.6.2 Termination location. A fill pipe shall terminate outside of a building at or above grade at a point not less than 2 feet (609.6 mm) from any building opening and not less than 5 feet (1524 mm) away from any subway grating at the same or lower level. A fill pipe shall terminate in a manner designed to minimize spilling when the filling hose is disconnected. Where No. 6 fuel oil is used, the fill pipe terminal shall be within 3 feet (914.4 mm) of the curb unless otherwise required by the Department of Transportation or the Transit Authority. If facilities exist for an oil delivery truck to drive onto the premises, the fuel-oil terminal may be located elsewhere other than the curb.

1305.6.3 Separate fill piping. Each storage tank shall be provided with a separate fill pipe, except that where a battery of tanks containing the same grade of oil is installed, a common fill and header pipe may be installed.

1305.6.4 Storage tank above the fill terminal. Where the top of the storage tank is above the fill pipe terminal, the fill pipe shall be connected to the top of the tank and provided with one of the following:

1. A shutoff valve and swing check valve located at the fill pipe terminal or installed in an accessible location inside the building at or below the level of the fill pipe terminal.
2. A dry disconnect fitting with a ball and a check valve located at the fill pipe terminal.

1305.6.5 Terminal opening. The fill opening shall be equipped with a tight metal cover designed to discourage tampering. All fill pipe terminals shall be of an approved type and shall be provided with lugs for embedding in concrete. In lieu of lugs, a set screw or threads to fasten the terminal to the fill pipe may be used. The outer flange of the fill pipe terminal or the seal cap shall be permanently marked: FUEL OIL. The fill pipe terminal shall be right-handed thread or provided with other equivalent means to receive the seal cap. The seal cap shall be suitably slotted for receiving an opening wrench, and an oilproof gasket inserted in a groove in the fill pipe terminal shall be provided so as to make the seal cap leakproof. A strainer shall not be required but if used, shall be of at least $\frac{1}{8}$ -inch (3.2 mm) mesh. Where a storage system for volatile flammable oil and a storage system for fuel oil are to be used in the same premises, the terminal of the volatile oil pipe shall be provided with a left-handed thread and the fill pipe fitting shall be of a different size than that required for the fill pipes to the tanks containing the volatile flammable oil.

1305.6.6 Spill containment. For fill terminal openings serving underground storage tanks greater than 110 gallons (416.4 L) and above-ground storage tanks greater than 660 gallons (2498.4 L), an approved overflow/spill containment device shall be provided.

1305.7 Normal vent piping. Normal vent piping shall comply with the requirements of Sections 1305.7.1 through 1305.7.9.

1305.7.1 Size. Normal vent sizes shall comply with the sizes listed in Tables 1305.7(1) and 1305.7(2); provided, however, for tanks other than those complying with the alternate tank design and construction standards contained in Section 1305.14, the normal vent shall not be smaller in size than the fill pipe.

TABLE 1305.7(1)
VENT PIPING FOR UNDERGROUND TANKS

TANK SIZE	MINIMUM VENT DIAMETER
660 gallons (2498 L) or less	1 $\frac{1}{4}$ inch (31.8 mm)

661 gallons (2505 L) to 3,000 gallons (11 355 L)	1½ inch (38.1 mm)
3,001 gallons (11 360 L) to 10,000 gallons (37 850 L)	2 inch (50.8 mm)
10,001 gallons (37 855 L) to 20,000 gallons (75 700 L)	2½ inch (63.5 mm)
Larger than 20,000 gallons (75 700 L)	3 inch (76.2 mm)

TABLE 1305.7(2)
NORMAL VENT PIPING FOR TANKS INSTALLED INSIDE BUILDINGS

TANK SIZE	MINIMUM VENT DIAMETER
660 gallons (2498 L) or less	1¼ inch (31.8 mm) ^a
Larger than 660 gallons (2498.4 L)	Sized to prevent abnormal pressure in the tank during filling but not smaller than the pipe size specified in Table 1305.7(1)

1305.7.2 Termination location. The location of the normal vent pipe terminations shall comply with the following:

1. Liquid fuel normal vent pipes shall terminate outside of buildings in a nonhazardous location at a point not less than 2 feet (609.6 mm) measured vertically or horizontally from any building opening and not less than 2 feet (609.6 mm) nor more than 12 feet (3657.6 mm) above the fill pipe terminal.
2. If the normal vent pipe terminal is not visible from the fill pipe terminal location, an electronic overfill alarm system or a 1-inch (25.4 mm) tell-tale line shall be connected to the tank and shall parallel the fill pipe and terminate at the fill pipe terminal with an unthreaded end. Such tell-tale lines shall be provided with a check valve set to prevent flow of surface water to the storage tank.
3. Normal vent pipes shall terminate sufficiently above the ground to avoid being obstructed with snow or ice.
4. Normal vent pipes from tanks containing heaters shall be extended to a location where oil vapors discharging from the normal vent will be readily diffused.

1305.7.3 Termination caps. Outer ends of normal vent pipes shall terminate in a weatherproof vent cap or fitting or be provided with a weatherproof hood. All normal vent caps shall have a minimum free open area equal to the cross-sectional area of the normal vent pipe and shall not employ screens finer than No. 4 mesh.

1305.7.4 Tank pressure. The tank shall be designed for the maximum static head that will be imposed with the normal vent piping filled with oil.

1305.7.5 Multiple tanks. A normal vent pipe shall be provided for each storage tank. Normal vent piping from multiple tanks of the same grade oil with not more than 660 gallons (2498.3 L) aggregate capacity may be combined. Where a battery of storage tanks complying with the alternate tank design and construction standards contained in Section 1305.14 designed to hold the same grade of oil with not more than 660 gallons (2498.3 L) aggregate capacity is installed, normal vent pipes may be run into a main header.

1305.7.6 Pitch. Normal vent pipes shall drain toward the tank. The normal vent pipes shall have no sags or traps where liquid can collect.

1305.7.7 Protection. Normal vent pipes shall be located so that they are not subjected to physical damage.

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1305.7.8 Cross-connection. Liquid fuel normal vent pipes shall not be cross-connected with fill pipes, supply lines to, or return lines from burners or other fuel-burning equipment.

1305.7.9 Tanks served by pumps. For tanks on any floor served by pumps, the normal vent shall be connected to the overflow pipe, shall be piped a minimum of 8 feet (2438.4 mm) above the overflow pipe and terminated directly to the outdoors. A liquid sensor shall be installed in the vent of the remotely served tank in order to shut down oil supply in the event of an overflow.

1305.8 Emergency relief vent piping. Each primary tank, the interstitial space of a secondary containment tank and each compartment of a compartment tank complying with UL 142 shall be provided with emergency relief venting. A tank's emergency relief vent piping and normal vent piping may be combined. The design and installation of the combined normal and emergency relief vent shall be in accordance with Section 1305.7 and Sections 1305.8.1 through 1305.8.4. Tanks designed in accordance with Section 1305.14 shall not require emergency relief vent piping. The use of a self-closing manway cover, a manway cover provided with long bolts that permit the cover to lift under internal pressure, or other type of emergency relief vent device inside the building, shall be prohibited.

1305.8.1 Piping. The combination normal and emergency relief venting shall be provided through an open vent pipe and shall be terminated outdoors.

1305.8.2 Reserved.

1305.8.3 Reserved.

1305.8.4 Capacity. The total relief venting capacity of the combination normal and emergency relief venting shall be in accordance with Section 22.7 of NFPA 30. Construction documents shall include calculations demonstrating that the extension of the normal and emergency vent piping is adequately sized to provide the required emergency vent flow while limiting the backpressure to less than the maximum pressure permitted by the design of the tank. Additionally, consistent with Section 27.7.3.4 of NFPA 30, where the design provides for a reduction in the required emergency relief venting capacity based upon the properties of the fuel oil to be stored in the tank, the construction documents shall include supporting calculations.

1305.9 Supplemental requirements for piping from transfer pumps to appliances or storage tanks above lowest floor. Fuel-oil piping systems utilizing pumps to transfer fuel oil to equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings shall comply with the requirements of Sections 1305.9.1 through 1305.9.12.

1305.9.1 Shaft enclosure. The piping from a transfer pump to equipment at levels above the lowest floor or storage tanks at levels above the lowest floor in buildings, the return or overflow piping, and vent piping where required to be connected to the main storage tank shall be enclosed in a shaft constructed of 4-inch (101.6 mm) concrete or masonry having a 4-inch (101.6 mm) clearance from all pipe or pipe covering in accordance with the *New York City Building Code*. Multiple fuel oil risers serving multiple systems are permitted within a common shaft enclosure. All shaft penetrations shall be fire stopped.

Exceptions:

1. An enclosure shall not be required for piping located within the same room as the pump, tank, or equipment where such room is itself enclosed with construction and materials having at least a 2-hour fire-resistance rating.
2. An enclosure shall not be required for outdoor piping in existing buildings in compliance with Section 1305.9.13.
3. A concrete or masonry shaft enclosure shall not be required for metallic double wall flexible fuel-oil piping installed in a shaft enclosure having a fire-resistance rating of at least 2-hours and that complies with Sections 703.2 and 708 of the *New York City Building Code*.

1305.9.2 Ducts or other piping in shafts. Pipe shafts containing fuel-oil piping shall not be penetrated by or contain other piping or ducts.

1305.9.3 Horizontal offsets. Where it is necessary to make horizontal offsets in the fuel-oil piping and pipe shafts, including the lowest level, such piping shall be enclosed in a fully welded outer containment of at least No. 10 standard Gage steel.

Exceptions:

1. The containment enclosure may be omitted on horizontal piping along the roof for a maximum length of 72 inches (1828.8 mm) from the enclosure, tank or the equipment served for hard piped connections and valves as required.
2. The containment enclosure may be omitted where a double metallic wall flexible fuel-oil piping system is installed as a horizontal offset, such pipe system does not need to be enclosed in an additional containment enclosure.

1305.9.3.1 Drainage and leak detection. Horizontal offsets shall be arranged to drain by gravity toward the tank room on the lowest level. Where the length of the horizontal offset exceeds 25 feet (7620 mm), the containment enclosure shall be constructed with an integral leakage collection reservoir of a cross-section area equal to the containment enclosure and at a minimum 6 inches (152.4 mm) deep. The reservoir shall be provided with a leak detection alarm. Where provided, all leak detection alarms shall be connected to a local audible alarm and to a remote alarm located at a supervising station. The wiring shall comply with the *New York City Electrical Code*.

Exception: A leak collection reservoir is not required where double metallic wall flexible fuel-oil piping is installed as a horizontal offset in accordance with Section 1302.9.

1305.9.3.2 Size. The containment enclosure shall be two sizes larger than the enclosed pipe for single pipe enclosures. Enclosures containing multiple pipes shall be sized to provide a minimum 1/2-inch (12.7 mm) clearance around the pipe bundle.

1305.9.3.3 Fire-resistance rating. In addition to the outer containment enclosure, horizontal piping offsets shall be further enclosed in construction having a 2-hour fire-resistance rating.

Exceptions:

1. A containment enclosure and 2-hour fire-resistance-rated enclosure are not required for the pipes serving the equipment or appliance within the room containing the pump, tank, or fuel-burning appliance where such room is itself enclosed with construction and materials having at least a 2-hour fire-resistance-rating.
2. A 2-hour fire-resistance-rated enclosure is not required for horizontal piping offsets located outside the building or on the roof.

1305.9.3.4 Horizontal offsets on roofs. Horizontal offsets on roofs shall be limited to a maximum vertical rise of one story.

1305.9.4 Drain pan at base of shafts with leak detection. A metal drain pan shall be installed at the base of every vertical shaft enclosing the fuel-oil piping. Where the bottom of the shaft is located outside the fuel oil storage tank room, the drain pan shall be connected to and drained into a containment enclosure at the horizontal offset of fuel-oil piping. On the lowest level in the building, the outer containment of horizontal offsets shall be terminated inside the main storage tank room or transfer pump room. A drain pipe shall be provided from the lowest point in the containment enclosure to a minimum 55-gallon (208.2 L) container with a leak detection alarm, arranged so as to sound an alarm and stop the transfer pump.

1305.9.5 Piping materials. Oil lines for appliances, equipment or tanks shall be steel pipe ASTM A 53/A 53M or ASTM A 106, grade B seamless Schedule 40 with welded connections up to the oil tank or equipment, except that fittings at the tank or equipment, shutoff valves and other fuel-oil flow and control devices may be screwed or flanged.

Exception: Flexible fuel-oil piping systems with continuous vacuum leak detection in accordance with Section 1302.9.

1305.9.6 Expansion. Provision shall be made for expansion in piping without the use of expansion joints.

1305.9.7 Movement and vibration. The piping shall be located and secured from movement so as to prevent undue stress on the piping and to isolate the piping from vibrations from any equipment.

1305.9.8 Connections to header. Pipe connections to the main header (supply or return) shall be made from the top of the header.

Exception: Systems with appliances above the lowest floor where such appliances are designed to operate utilizing fuel pumped as needed from the lowest floor and without utilizing fuel oil stored via header above the lowest floor.

1305.9.9 Vacuum breaking devices. Vacuum breaking devices shall be designed for their required use. Vacuum breaking devices shall be hard-piped to a reservoir with a leak detection alarm, arranged so as to sound an alarm. The alarm shall be connected to a local audible alarm and to a remote alarm located at a supervising station. The wiring shall comply with the *New York City Electrical Code*.

1305.9.10 Reserved.

1305.9.11 Pipe size; fuel storage tank above the lowest floor. In systems with appliances above the lowest floor where such appliances are designed to operate utilizing fuel stored above the lowest floor, piping diameters shall not exceed 4 inches (101.6 mm).

Exception: Where an applicant demonstrates, by calculations in the construction documents, that a greater diameter is necessary to ensure the proper flow for the functioning of the system.

1305.9.12 Pipe size; direct feed without fuel storage above the lowest floor. In systems with appliances above the lowest floor, where such appliances are designed to operate utilizing fuel pumped as needed from the lowest floor on demand sequenced with the equipment operation, piping diameters shall not exceed 4 inches (101.6 mm). Piping shall not be used for fuel storage purposes.

Exceptions:

1. Piping diameters may exceed 4 inches (101.6 mm) where an applicant demonstrates, by calculations in the construction documents, that a greater diameter is necessary to ensure the proper flow for the functioning of the system.
2. Piping diameters within rooms containing fuel-burning equipment may exceed the calculated design flow pipe size to provide limited storage to prime equipment, provided such storage is counted toward the maximum permitted oil storage per story, as provided for in Section 1305.11.1.3.

†††1305.9.13 Outdoor fuel-oil piping in existing buildings. Outdoor fuel-oil piping in existing buildings, including vertical piping, utilizing pumps to transfer fuel oil to appliances at levels above the lowest floor or to storage tanks at levels above the lowest floor in buildings shall additionally comply with the following:

1. Piping shall be located a minimum of 10 feet (3048 mm) from lot lines or a 2-hour fire-rated-enclosure shall be provided.
2. Piping shall be located a minimum of 3 feet (914.4 mm) from building openings and combustible construction.
3. Vertical outdoor fuel-oil piping shall be enclosed in a fully welded outer containment of at least No. 10 standard Gage steel sized in accordance with Section 1305.9.3.2.
4. Horizontal outdoor fuel-oil piping shall comply with Section 1305.9.3.
5. In addition to the requirement of Section 1305.9.4, a drain pipe shall also be provided from the lowest point in the vertical outdoor containment enclosure to a minimum 55-gallon (208.2 L) container with a leak detection alarm, arranged so as to sound an alarm and stop the transfer pump. The container may be located immediately inside the building.
6. Materials and supports directly exposed to the weather shall be stainless steel or other corrosion resistive material.
7. Details shall be provided for piping supports and connections to building structure.
8. Penetrations of building walls shall be encased in a protective pipe sleeve. The annular space between the piping and the sleeve shall be sealed in accordance with the *New York City Building Code*.
9. Piping shall be grounded in accordance with the *New York City Electrical Code* Section 250.104 (B).
10. Piping shall be protected from vehicle impact and physical damage.
11. Flexible fuel-oil piping systems other than in accordance with Section 1302.9 shall not be utilized.
12. Egress paths shall not be obstructed.

13. Roof access shall comply with Section 306.5.
14. Piping shall be identified by a permanent label or tag at intervals not more than 40 feet (12 192 mm) in length and at all changes of direction. The label or tag shall be located outside of the enclosure.

1305.10 Devices to control flow to oil-burning appliances including generators. The following requirements shall apply:

1. The pressure in oil lines to oil-burning appliances located above the lowest floor of a building shall not be more than is required to circulate oil to and from the burners, and all parts of the oil system shall be capable of withstanding the maximum working pressure in that part of the system.
2. A remote control shall be provided to stop the flow of oil to any burner wherever located, and to any oil-burning appliances located on levels above the lowest floor in buildings as well as the fuel oil transfer pumps. Such control shall be located outside the entrance to the room in which the burner is located and as close to such entrance as practicable, except that when an outside location is impracticable, such control may be located immediately inside the room in which the burner is located, provided such location is accessible at all times. All such controls shall be permanently labeled: "REMOTE CONTROL FOR BURNER," or as appropriate to the oil-burning appliance. On storage tanks of 60 gallons (227.1 L) or less capacity used with manually operated appliances, such remote control may be installed in the supply lines between tank and burner.
3. In systems where either steam or air is used for atomizing the oil, the oil and the atomizing supply shall be interlocked so that where the supply of either is interrupted, the supply of the other will be immediately cut off.

1305.11 Limitations on quantities of fuel-oil storage. Quantities of fuel-oil storage shall be limited in accordance with the provisions of this section. For the purposes of this section, fuel oil stored on roofs shall be deemed inside of buildings and located on the floor to which they are adjacent.

1305.11.1 Inside of buildings. A total of not more than 100,000 gallons (378 541.2 L) shall be stored inside of any building. Oil storage inside of buildings shall also comply with applicable requirements of Sections 1305.11.1.1 through 1305.11.1.3.

1305.11.1.1 Inside of buildings; below ground. The maximum size of each below-ground oil-storage tank inside of a building shall be 35,000 gallons (132 489.4 L).

1305.11.1.2 Inside of buildings; above ground on the lowest floor. Fuel-oil storage tanks installed above ground on the lowest floor of a building shall be mounted on and anchored by adequate noncombustible supports. The maximum size of each individual tank shall be 660 gallons (2498.4 L), and a total of not more than 1,375 gallons (5204.9 L) shall be stored within the same 2-hour fire and smoke barrier construction.

Exceptions: Fuel-oil storage tanks shall be permitted to exceed 660 gallons (2498.4 L), and the total quantity within a fire area shall be permitted to exceed 1,375 gallons (5204.9 L) in accordance with any one of the following options:

1. **Buildings of Type I, II, IIIA, IV or VA construction with a total limit of 15,000 gallons (56 781.1 L).** The maximum size of each individual tank shall be 15,000 gallons (56 781.1 L) provided that all such tanks are located in a room or enclosure dedicated to oil storage that is separated from the rest of the building by fire and smoke barrier construction of at least 3 hours. Notwithstanding Section 1305.11.1, in such cases, the maximum total quantity in the building shall be limited to 15,000 gallons (56 781.1 L).
2. **Buildings of Type IIIB or VB construction with a total limit of 10,000 gallons (37 854.1 L).** The maximum size of each individual tank shall be 10,000 gallons (37 854.1 L) provided that all such tanks are located in a room or enclosure dedicated to oil storage that is separated from the rest of the building by fire and smoke barrier construction of at least 3 hours. Notwithstanding Section 1305.11.1, in such cases, the maximum total quantity in the building shall be limited to 10,000 gallons (37 854.1 L).
3. **Buildings of any type construction with a total limit of 100,000 gallons (378 541.8 L).** The maximum size of each individual tank shall be 25,000 gallons (94 635.3 L) provided that all such tanks are enclosed in a vault (i) with walls, floor, and top having a fire and smoke barrier construction of not less than 3 hours, (ii) with such walls bonded to the floor, and (iii) with such top and walls of the vault independent of the building structure. An exterior building wall having a fire and smoke barrier

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construction of not less than 3 hours shall be permitted to serve as a wall of the vault. The vault shall be located in a dedicated room or area of the building that is cut off vertically and horizontally from other areas and floors of the building by assemblies having a fire and smoke barrier construction of not less than 2 hours. Where the aggregate fuel-oil storage on the lowest level of the building exceeds 50,000 gallons (189 270.6 L), such storage shall be protected with an alternate extinguishing system complying with the *New York City Fire Code* and Section 904 of the *New York City Building Code*.

1305.11.1.3 Inside of buildings; above the lowest floor. Fuel oil above the lowest floor inside of a building shall be limited to 330 gallons (1249.1 L) per story. The maximum quantity shall include oversized piping as described in Section 1305.9.12. Piping installations shall comply with the requirements of Section 1305.9.

Exception: Fuel-oil storage capacity in special flood hazard areas and shaded X-Zones, as defined in Chapter 2 of the *New York City Building Code*, shall comply with Section G307.4 of Appendix G of the *New York City Building Code*.

1305.11.2 Outside of buildings. Oil storage outside of buildings shall comply with applicable requirements of Sections 1305.11.2.1 and 1305.11.2.2.

1305.11.2.1 Outside of buildings; below ground. The maximum size of each below-ground oil-storage tank outside of a building shall be 35,000 gallons (132 489.4 L).

1305.11.2.2 Outside of buildings; above ground. The maximum size of each above-ground oil-storage tank outside of a building shall be 100,000 gallons (378 541.2 L).

1305.12 Standards for tank design. Tanks shall be designed and constructed in compliance with Sections 1305.12.1 and 1305.12.2.

1305.12.1 Below ground. Tanks located below ground, inside or outside of buildings, shall comply with any one of the following design standards, as appropriate for the specific installation as determined by the engineer. Cathodic protection shall be provided in accordance with subpart 613-2 of the *New York State Department of Environmental Conservation's Petroleum Bulk Storage Code*.

1. UL 58; such tanks shall be listed and labeled;
2. UL 1316; such tanks shall be listed and labeled; or
3. Alternate tank design and construction standards contained in Section 1305.14.

†††**1305.12.2 Above ground.** Tanks located above ground, inside or outside of buildings, shall comply with any one of the following design standards, as appropriate for the specific installation as determined by the engineer:

1. UL 80; such tanks shall be listed and labeled;
2. UL 142; such tanks shall be listed and labeled;
3. UL 2258; such tanks shall be installed only in one- or two-family dwellings in accordance with Section 1305.15 and shall be listed and labeled;
4. ASME *Boiler and Pressure Vessel Code*, Section VIII, Division 1 or 2; such tanks shall be labeled; or
5. Alternate tank design and construction standards contained in Section 1305.14.

1305.13 Installation of tanks. Tanks shall be installed in accordance with the provisions of Sections 1305.13.1 through 1305.13.4. For the purposes of this section, fuel oil stored on roofs shall be deemed inside of buildings. At least 15 inches (381 mm) clearance shall be provided over tanks and on all sides between the tanks and the enclosure, embankment or dike or greater clearance as required by the manufacturer's instructions.

1305.13.1 Below ground. Tanks located below ground, inside or outside of buildings, shall comply with the following requirements:

1. **Containment.** Fuel-oil tanks having a capacity of more than 660 gallons (2498.4 L) shall be provided with secondary containment intended to prevent any leakage of fuel oil from the tank from entering the environment. The capacity of the containment shall equal or exceed the capacity of the tank served.

2. **Burial.** Regardless of capacity, fuel-oil tanks shall be buried with the top of the tank at least 2 feet (609.6 mm) below ground. Tanks shall be placed in firm soil and shall be surrounded by clean sand or well-tamped earth, free from ashes or other corrosive substance, and free from stones that will not pass a 1-inch (25.4 mm) mesh.
3. **Anchorage.** When necessary to prevent floating, fuel-oil tanks, regardless of capacity, shall be securely anchored.
4. **Distance to foundations.** Regardless of capacity, no fuel-oil tank shall be buried within 3 feet (914.4 mm) of any foundation wall or footing.
5. **Special limitations near subways.** Regardless of capacity, no fuel-oil tank shall be placed within 20 feet (6096 mm) of the outside line of a subway wall. For the purpose of the foregoing requirement, a subway shall be deemed to include any subsurface railroad or rapid transit roadbed.

1305.13.2 Above ground; on the lowest floor inside a building. Tanks located above ground, on the lowest floor inside of buildings, shall comply with the following requirements:

1. **Enclosure of room.** Installation of tank(s) and enclosure of room shall comply with Section 1305.11.1.2.
2. **Fire-extinguishing system.** Fire extinguishing systems shall comply with Section 1305.11.1.2.
3. **Ventilation.** Rooms containing fuel-oil tanks shall be ventilated to limit the concentration of vapors within the room at or below 25 percent of the Lower Flammable Limit (LFL) of the fuel oil being used.
4. **Containment.** Fuel-oil tanks having a capacity of more than 660 gallons (2498.4 L) storage shall be provided with secondary containment intended to capture any leakage of fuel oil from the tank. The capacity of the containment shall equal or exceed the capacity of the tank served. For fuel storage, the capacity shall include the portion of the tank up to the height of the containment.
5. **Special limitations near subways.** Regardless of capacity, fuel-oil tanks located within the outer lines of the subway, or within 20 feet (6096 mm) of the outside line of a subway shall be placed within a welded steel oil-tight pan of not less than No. 18 Gage metal suitably reinforced and of capacity to contain the contents of the tank. For the purpose of the foregoing requirement, a subway shall be deemed to include any subsurface railroad or rapid transit roadbed.

1305.13.3 Above ground; above the lowest floor inside a building. Regardless of capacity, fuel-oil tanks and fuel-oil-burning appliances located above ground, above the lowest floor inside of buildings, shall comply with the following requirements:

1. **Enclosure of room.** Fuel-oil tanks and fuel-oil-burning appliances shall be located in a dedicated room or enclosure, having a fire-resistance rating of at least 2 hours. Rooftop tanks need not be enclosed provided that all exterior walls and roofs within 10 feet (3048 mm) horizontally and 20 feet (6096 mm) vertically have a fire-resistance rating of at least 2 hours or are otherwise separated by a constructed assembly with a fire-resistance rating of at least 2 hours.
2. **Fire-extinguishing system.** Rooms containing fuel-oil tanks and fuel-oil-burning appliances shall be equipped with an automatic sprinkler system in accordance with Section 903.3.1 of the *New York City Building Code*. To prevent overfilling of the containment barriers, sprinkler shut-offs shall be located on the outside of tank and generator rooms and prominently placarded for immediate control by the Fire Department.
3. **Smoke detection.** Rooms containing fuel-oil tanks and fuel-oil-burning appliances shall be equipped with automatic smoke detection in accordance with Section 907 of the *New York City Building Code*, except that heat detectors may be utilized where, during normal operation, products of combustion are present in sufficient quantity to actuate a smoke detector.
4. **Ventilation.** Rooms containing fuel-oil tanks shall be ventilated to limit the concentration of vapors within the room at or below 25 percent of the Lower Flammable Limit (LFL) of the fuel oil being used.
5. **Containment.** Fuel-oil tanks, fuel-oil-burning appliances, and related equipment shall be provided with secondary containment area intended to capture any leakage of fuel oil. Floor drains shall be prohibited in

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containment areas. For tanks, the capacity of the containment area shall equal or exceed 2 times the capacity of the tank served. For fuel storage, the capacity shall include the portion of the tank up to the height of the containment. For fuel-oil-burning appliances and related equipment, the capacity of the containment area shall equal or exceed 1.5 times the fuel capacity of the equipment or appliance.

6. **Transfer pumps.** Fuel-oil tanks shall be filled by means of a transfer pump supplied from a primary storage tank located on the lowest floor. A separate transfer pump and piping circuit shall be provided for each storage tank installed above the lowest floor. No intermediate pumping stations shall be provided between the storage tank and the transfer pump. Appropriate devices shall be provided for the automatic and manual starting and stopping of the transfer pumps so as to prevent the overflow of oil from these storage tanks. The transfer pumps shall be located above the tank secondary containment level or outside the tank containment area.
7. **Indicators and alarms.** Indicators and alarms shall be provided for fuel-oil tanks and rooms containing fuel-oil-burning appliances, including a level sensor for height and capacity of fuel oil, high and low levels, and leak detection. The float switch shall be provided within the containment areas and shall be arranged so as to sound an alarm and stop the transfer pump in case of failure of the tank or the control in the tank. These indicators shall be connected to a local audible alarm in the tank room and to a remote alarm located at a supervising station. The wiring shall comply with the *New York City Electrical Code*.
8. **Weekly testing.** The operation of the float switch shall be tested at least once each week by the holder of the certificate of fitness as provided for in Sections 1301.10 and 1308.3.

1305.13.4 Above ground; outside a building. Tanks located above ground, outside of buildings, shall comply with the following requirements:

1. **Containment.** Regardless of capacity, each fuel-oil storage tank shall be protected by an embankment or dike. Such protection shall have a capacity at least $1\frac{1}{2}$ times the capacity of the tank so surrounded (including the portion of the tank up to the height of the containment) and shall be at least 4 feet (1219.2 mm) high, but in no case shall the protection be higher than one-quarter the height of the tank when the height of the tank exceeds 16 feet (4876.8 mm). Embankments or dikes shall be made of earthwork with clay core, of masonry, of reinforced concrete or of steel. Earthwork embankments shall be firmly and compactly built of good earth free from stones, vegetable matter, or other similar material, and shall have a flat section of at least 3 feet (914.4 mm) at the top and a slope of at least $1\frac{1}{2}$ feet (457.2 mm) rise to 2 feet (609.6 mm) of run on all sides. Concrete, masonry or steel dikes shall be designed so as to contain safely all of the oil in the tank so surrounded. Embankments or dikes shall be continuous and unpierced, and the outside toe shall be located at least 5 feet (1524 mm) inside of the property line, and no less than 5 feet (1524 mm) from a driveway or parking area.
2. **Distances to buildings, lot lines and other tanks.** Fuel oil storage tanks shall be located in accordance with the following:
 - 2.1. Storage tanks of a capacity greater than 330 gallons (1249.2 L) shall be not less than $1\frac{1}{4}$ tank diameters and in no case less than 10 feet (3048 mm) from the tax lot line, the nearest building not served by such tanks or other fuel oil storage tank. However, in no case shall the clearance between individual tanks and the tax lot line be less than the distance fixed by the following formula:
$$M.C. = 10 + 4 ((G - 330) / 5000) \quad \text{(Equation 13-1)}$$
where:
$$M.C. = \text{Minimum clearance from nearest surface of tank to tax lot line, in feet.}$$
$$G = \text{Capacity of tank, in gallons.}$$
 - 2.2. Storage tanks with a capacity of not more than 330 gallons (1249.2 L) shall be not less than 5 feet (1524 mm) from the tax lot line, the nearest neighboring building or other fuel oil storage tank.
3. **Means of egress.** Tanks shall be located so as not to obstruct or interfere with any means of egress.

1305.14 Alternate tank design and construction standards. Oil-storage tanks, other than those conforming to ASME *Boiler and Pressure Vessel Code*, Section VIII, Division 1 or 2, UL 58, UL 80, UL 142, or UL 1316 shall be designed, constructed and installed in accordance with the requirements of Sections 1305.14.1 through 1305.14.5.

1305.14.1 General construction standards. All tanks shall comply with the requirements of Sections 1305.14.1.1 through 1305.14.1.9.

1305.14.1.1 Materials and workmanship. All fuel-oil storage tanks shall be built of steel plates or sheets, made by the open hearth or basic oxygen process. Such steel shall be free from physical imperfections, and shall be new, in good condition and free from rust.

1305.14.1.2 Assembly. Tanks, flanges or other pipe connections shall be welded. Filler of any kind between plates shall be prohibited.

1305.14.1.3 Corrosion resistance. Tanks to be buried shall be cleaned and then coated on the outside with two coats of corrosion protective material. They shall be further protected by a coating of hot tar, asphalt, or equivalent rust-resistive material, applied at the work site. Tanks installed inside buildings above ground shall be coated with one coat of corrosion protective material.

1305.14.1.4 External loads on underground tanks. All buried storage tanks shall be constructed of at least $\frac{1}{4}$ -inch-thick (6.4 mm) metal and shall be designed to withstand any external loads to which the tank may be subjected.

1305.14.1.5 Identification. At the time of installation, all storage tanks shall bear a permanently fixed plate, spot welded or equivalent, bearing the name of the tank manufacturer, the gage of the material, and capacity of the tank. Shop-fabricated storage tanks shall be installed without structural alteration.

1305.14.1.6 Openings. All openings shall be through the top of the storage tank, except that storage tanks of 330 gallon (1249.2 L) capacity or less, located above ground on the lowest story, may be provided with a $\frac{1}{2}$ -inch (12.7 mm) opening for gravity discharge in the bottom of the storage tank.

1305.14.1.7 Manholes. Tanks for No. 1, No. 2, No. 3 and No. 4 commercial-grade oils need not have manholes. However, if manholes are used for tanks containing such oils, the manhole covers shall be bolted and made gas tight. Tanks for No. 5 and No. 6 commercial-grade oils shall have manhole covers bolted or otherwise secured to the tanks and kept hydrostatically tight at all times. Tanks 330 gallon (1249.2 L) capacity or less, and all other tanks without manholes, shall be provided with a two screwed connection on the top of the tank to permit measuring the level of the oil within.

1305.14.1.8 Electrical grounding. Tanks outside of buildings shall be electrically grounded in accordance with the requirements for equipment grounding of the *New York City Electrical Code*.

1305.14.1.9 Protection from heat and flame. Tanks shall be located at least 7 feet (2133.6 mm), measured in the most direct manner, from any source of exposed flame unless protected as provided in Section 1305.11.1.2, Exception 3, and at least 2 feet (609.6 mm) from any surface where the temperature exceeds 165°F (73.9°C).

1305.14.2 Additional construction standards for cylindrical tanks exceeding 330 gallons (1249.2 L). Cylindrical tanks, including oval, elongated oval, or round tanks, exceeding 330 gallons (1249.2 L) shall comply with the requirements of Sections 1305.14.2.1 through 1305.14.2.3.

Exception: Such above-ground vertical tanks that are outside of buildings shall comply with Sections 1305.14.1 and 1305.14.5.

1305.14.2.1 Thickness. The minimum thickness shall be as follows:

1. Tanks 36 inches (914.4 mm) in diameter or less shall have at least a $\frac{1}{4}$ -inch (6.4 mm) shell and $\frac{1}{4}$ -inch (6.4 mm) heads.
2. Tanks 37 inches (940 mm) to 72 inches (1828.8 mm) in diameter shall have at least a $\frac{1}{4}$ -inch (6.4 mm) shell and $\frac{5}{16}$ -inch (7.9 mm) heads.
3. Tanks 73 inches (1854 mm) to 120 inches (3048 mm) in diameter shall have at least a $\frac{5}{16}$ -inch (7.9 mm) shell and $\frac{3}{8}$ -inch (9.5 mm) heads.
4. Tanks over 120 inches (3048 mm) in diameter shall be of at least $\frac{3}{8}$ -inch (9.5 mm) steel and shall be stiffened by angle rings or equivalent members so as to retain their cylindrical form.

1305.14.2.2 Dished heads. Dished heads for such tanks shall have a curvature the radius of which is not greater than the diameter of the tank. Dished heads shall be formed with an adequate cylindrical extension rim to provide a welding surface.

1305.14.2.3 Flat heads. If flat heads are used, they shall be braced in the same manner as described for the bracing of flat sides of rectangular tanks as provided for in Section 1305.14.3.

1305.14.3 Additional construction standards for rectangular tanks exceeding 330 gallon (1249.2 L). Rectangular tanks exceeding 330-gallon (1249.2 L) capacity shall comply with the requirements of Sections 1305.14.3.1 through 1305.14.3.6.

1305.14.3.1 Thickness. Plates for rectangular tanks of more than 330 gallon (1249.2 L) capacity shall be at least $\frac{5}{16}$ inch (7.9 mm) thick.

1305.14.3.2 Corners. Corners may be made up by bending the plates or by using angles.

1305.14.3.3 Seams. All tanks shall have full penetration $\frac{5}{16}$ welds at all seams.

1305.14.3.4 Bracing. All flat surfaces of rectangular tanks shall be braced by structural members or rods.

1305.14.3.5 Structural work. All structural members shall be designed in accordance with the requirements of the *New York City Building Code*.

1305.14.3.6 Connections. Connections between bracing members and the sides of the tank shall be designed so that the connection will not fail before the member will fail.

1305.14.4 Additional construction standards for tanks 330 gallons (1249.2 L) or less. Storage tanks with a capacity of less than or equal to 330 gallons (1249.2 L) shall have a minimum thickness of shell and head plates of No. 10 manufacturer's standard Gage steel plate. Storage tanks of 60 gallon (227.1 L) capacity or less shall be similarly constructed but need not be thicker than No.14 manufacturer's standard Gage.

Exceptions:

1. Such vertical above-ground cylindrical tanks outside of buildings shall comply with Sections 1305.14.1 and 1305.14.5.
2. Such underground tanks need comply only with Section 1305.14.1.
3. Storage containers of 6 gallons (22.7 L) or less used with burners or oil-burning heaters need only be designed so as to withstand a hydrostatic pressure test of at least 5 psi (34.5 kPa) without permanent deformation, rupture, or leakage, and shall be approved. Such containers shall be installed with rigid metal fasteners for wall, floor, or stand-type installations, and shall be protected against mechanical damage. Portable storage containers of 6 gallons (22.7 L) or less may be filled by a pump mounted on a storage tank, provided that the pump is approved.

1305.14.5 Additional construction standards for vertical above-ground cylindrical tanks outside of buildings exceeding 6 gallons (22.7 L). Vertical above-ground cylindrical storage tanks exceeding 6 gallons (22.7 L) that are located outside of buildings shall comply with the requirements of Sections 1305.14.5.1 through 1305.14.5.3.

1305.14.5.1 Plates. Such tanks shall be built of steel plates of the quality required for cylindrical tanks in accordance with Section 1305.14.2.

1305.14.5.2 Thickness. The minimum thickness of shell or bottom plates shall be $\frac{1}{4}$ inch (6.4 mm), and the minimum thickness of roof plates $\frac{1}{8}$ inch (3.2 mm). The thickness of shell plates shall be determined in accordance with the following formula:

$$t = \frac{P \times R \times F}{T \times E} \quad \text{(Equation 13-2)}$$

where:

t = Thickness of shell plate in inches.

P = Head pressure at bottom of ring under consideration in psi.

R = Radius of shell, in inches.

F = Factor of safety (taken as 5).

T = Tensile strength of plate, in psi, as verified by mill test certificate.

E = Efficiency of vertical joint in ring under consideration. E shall in no case be taken greater than 1.00.

1305.14.5.3 Seams. Roof plates shall have welded water-tight seams, and the roof shall be built to shed water. Bottom plates shall have welded seams. Shell plate seams shall be designed to develop the full strength of the plate.

1305.15 Above-ground nonmetallic fuel-oil storage tanks for one- and two-family dwellings. Above-ground non-metallic fuel-oil storage tanks may be used for indoor and outdoor storage and supply of fuel-oil in accordance with this section.

1305.15.1 Use. Nonmetallic tanks shall be permitted for one- and two-family dwellings only. Use of nonmetallic tanks in multi-family dwellings is prohibited. Nonmetallic fuel-oil tanks shall not be used for storage of waste-oil and flammable liquids.

1305.15.2 Capacity. The capacity of nonmetallic fuel-oil storage tanks shall be not less than 60 gallons (227.1 L) and not more than 660 gallons (2498.4 L).

1305.15.3 Construction standards. Nonmetallic tanks shall be constructed per Sections 7, 8, 9, 10, 11 and 12 of UL 2258 and Chapter 7 of NFPA 31 and shall consist of a non-metallic primary tank and a metallic secondary tank.

1305.15.4 Identification. Above-ground nonmetallic fuel-oil storage tanks shall be marked in accordance with Sections 40, 41 and 42 of UL 2258. The label shall also include the biodiesel percentage as identified in the scope of UL 2258.

1305.15.5 Location. Above-ground nonmetallic fuel-oil storage tanks installed indoors shall not be exposed to sunlight on any plastic parts. Above-ground nonmetallic fuel-oil storage tanks installed outdoors shall be assembled with a required cover per the manufacturer's recommendations. Outdoor installations shall comply with Section 1305.13.4.

1305.15.6 Installation. Nonmetallic fuel-oil tanks shall be installed in accordance with the manufacturer's instructions, this code and NFPA 31. Installation of above-ground nonmetallic fuel-oil storage tanks shall be performed by an oil-burning equipment installer licensed by the department with training and experience certified by the manufacturer.

1305.15.7 Testing. Nonmetallic fuel-oil tanks shall be tested in accordance with UL 2258.

1305.16 Method of ventilation of fuel oil storage tank rooms. Where required by this code, fuel oil storage tank rooms shall be ventilated to limit the concentration of vapors within the room at or below 25 percent of the Lower Flammable Limit (LFL) of the fuel oil being used via an independent exhaust system to the outdoors complying with the following:

1. The exhaust system shall operate continuously at a rate of two air changes per hour.

Exception: Continuous operation shall not be required where listed detection devices are interlocked to operate such exhaust system to maintain concentration of vapors below 25 percent of LFL.

2. Multiple tank rooms may be served by a shared exhaust system.
3. Make-up air shall be provided via a ducted building ventilation system.
4. All dampers, including smoke dampers, shall be interlocked with the exhaust air system.

1305.16.1 Replacement of tanks in prior code buildings. Replacement of tanks in prior code buildings may utilize an existing gravity ventilation system complying with the following:

1. The total net free area of supply and exhaust openings shall be equal to at least 1 percent of the floor area of the room, equally divided between the supply and exhaust; and
2. Independent supply and exhaust openings to the outside; or
3. Independent supply and exhaust ducts to the outside.

SECTION MC 1306 OIL GAUGING

FUEL-OIL PIPING AND STORAGE

1306.1 Level indication. All tanks located inside buildings shall be equipped with a method of determining the oil level.

1306.2 Test wells. Test wells shall not be installed inside buildings. Unused tank openings shall be permanently sealed. For outside service, test wells shall be equipped with a tight metal cover designed to discourage tampering.

1306.3 Inside tanks. The gauging of inside tanks by means of measuring sticks shall not be permitted. An inside tank provided with fill and vent pipes shall be provided with a device to indicate either visually or audibly at the fill point when the oil in the tank has reached a predetermined safe level.

1306.4 Gauging devices. Gauging devices such as liquid level indicators or signals shall be designed and installed so that oil vapor will not be discharged into a building from the liquid fuel supply system.

1306.5 Gauge glass. A tank used in connection with any oil burner shall not be equipped with a glass gauge or any gauge which, when broken, will permit the escape of oil from the tank.

1306.6 Storage above lowest floors. Fuel oil stored above the lowest floor shall also comply with the indicator requirements of Section 1305.13.3, Item 7.

SECTION MC 1307 FUEL-OIL VALVES

1307.1 Building shutoff. For outside or below ground tanks, a shutoff valve shall be installed on the fuel-oil supply line at the entrance to the building. Inside or above-ground tanks shall have valves installed at the tank. The valve shall be capable of stopping the flow of fuel oil to the building or to the appliance served where the valve is installed at a tank inside the building. Valves shall comply with UL 842.

1307.2 Appliance shutoff. A shutoff valve shall be installed at the supply connection to each appliance and a check valve installed at the return connection.

1307.3 Pump relief valve. A relief valve shall be installed on the pump discharge line where a valve is located downstream of the pump.

1307.4 Fuel-oil heater relief valve. A relief valve shall be installed on the discharge line of fuel-oil-heating appliances.

1307.5 Relief valve operation. Relief valves shall be set to discharge at not more than $1\frac{1}{2}$ times the maximum working pressure of the system. The discharge from relief valves shall be returned to the storage tank or to the supply line. Shutoff valves are not permitted in the line of relief.

SECTION MC 1308 TESTING

1308.1 Testing required. Fuel-oil storage tanks, other than tanks complying with the alternate tank design and construction standards contained in Section 1305.14, shall be tested in accordance with NFPA 31.

1308.2 Hydrostatic test. All liquid-fuel piping, and all tanks complying with the alternate tank design and construction standards contained in Section 1305.14, shall be hydrostatically tested for tightness by the contractor who made the installation before the work is closed in and before the system is operated. The piping shall be tested at $1\frac{1}{2}$ times the maximum working pressure applicable to that part of the piping system but at a pressure not less than the test pressure required for the storage tank. The minimum pressure for testing tanks shall be $1\frac{1}{2}$ times the maximum working pressure applicable to the tank but in no case less than 25 psig (172.4 kPag), except as provided for containers 6 gallons (22.7 L) or less in capacity as provided for in Section 1305.14.4, Exception 3. The hydrostatic pressure shall be maintained until all joints and connections have been visually inspected for leaks, but in no case for less than $\frac{1}{2}$ hour. The tank shall not show any permanent deformation as a result of the test. A record shall be kept of the pressure tests showing the name of the contractor and the pressures at which the piping and the tank were tested.

1308.3. Weekly testing. For fuel-oil tanks and fuel-oil-burning equipment located above ground, above the lowest floor inside of buildings, the operation of the float switch shall be tested at least once each week by the holder of the certificate of fitness as provided for in Sections 1301.9 and 1305.13.3, Item 8.